NITESH KUMAR MISHRA

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SUMMARY

AI/ML Engineer building ML models (XGBoost, YOLO) and LLM agents (LangGraph, RAG, MCP) with FastAPI.

EXPERIENCE

AI/Backend Engineering Intern – Techvruk
Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, AI/Computer Vision

- Designed a **microservices architecture** of 6 services for an AI-powered circuit diagram analysis platform.
- Performed **manual annotation of 30+ P&ID diagrams** for YOLO-based symbol detection and model training.
- Built **15+ FastAPI REST APIs** for diagram upload, processing, authentication, and validation.
- Designed **PostgreSQL/SQLAlchemy** models and Alembic migrations for companies, users, roles, and diagrams.

PROJECTS

TripMate AI – Multi-Agent Travel Planner Live | GitHub

Tech Stack: FastAPI, LangGraph, Groq, PostgreSQL, MCP

- Built a LangGraph-based **multi-agent travel planner** with supervisor routing and **4+ specialized agents** (Flight, Hotel, Weather, and Budget), incorporating guardrails and Human-in-the-Loop approval.
- Integrated a **LiteLLM gateway with Groq-to-Gemini fallback** and **3 MCP tools** (AviationStack, Tavily, and OpenWeather) to retrieve real-time travel data.
- Developed an **automated evaluation framework** with JSON reports and unit tests covering agent routing, guardrails, and flight-provider behavior, achieving **5/5 passing baseline test cases**.

Credit Risk Assessment Live | GitHub

Tech Stack: Python, XGBoost, scikit-learn, SHAP, Pandas

- Built an end-to-end **loan default prediction** pipeline on 32K records (22% default rate) with leakage-safe scikit-learn **Pipeline/ColumnTransformer**; handled 3.6:1 imbalance via class weighting.
- XGBoost beat a Logistic Regression baseline in 5-fold CV (**ROC-AUC 0.94 vs 0.87**); **RandomizedSearchCV** tuning (150 configs) lifted test **F1 from 0.65 to 0.81**.
- Used **SHAP** global and per-applicant explanations to show loan-to-income ratio, income, interest rate, and loan grade drive default risk; applied Platt scaling for calibration.

CivicAI – AI-Powered Municipal Grievance Platform Live | GitHub

Tech Stack: Python, YOLO11, FastAPI, React, Multimodal AI

- Trained a custom **YOLO11n** detector for 4 civic issue classes at 640px (**0.59 mAP@50, 0.65 precision** on validation) as the perception layer of a hybrid CV + multimodal AI pipeline.
- Accepted detections at $\geq$0.60 confidence; others fall back to a multimodal LLM for severity and **safety risk**.
- Built deterministic authority routing, severity-aware **SLA deadlines** (24–336 h), and automated email escalation.

Cashora – Personal Finance Platform Live | GitHub

Tech Stack: Python, FastAPI, React, PostgreSQL, LangGraph, Groq, FAISS

- Built a bank-statement analysis platform (**React, FastAPI**) with HDFC and PNB parsers; extracted **678/678 transactions** from a 7-month HDFC statement, matching the bank's own count.
- Developed an **agentic AI assistant** (LangGraph, Groq, FAISS RAG) with 7 deterministic financial tools so totals are computed by tools, not the LLM, with streamed responses.
- Added **OCR** for scanned statements (13/13 transactions recovered) and rule-based categorization into 10 categories.

SKILLS

Languages: Python, SQL, C++, JavaScript, TypeScript
Machine Learning: scikit-learn, XGBoost, SHAP, Feature Engineering, Hyperparameter Tuning, Cross-Validation
AI/GenAI: LLMs, RAG, LangChain, LangGraph, AI Agents, MCP, Vector Databases, YOLO, OCR
Data & Backend: Pandas, NumPy, Matplotlib, FastAPI, REST APIs, SQLAlchemy, PostgreSQL, MongoDB
Cloud & DevOps: Docker, AWS EC2, Nginx, GitHub Actions, Linux

EDUCATION

Maharaja Surajmal Institute of Technology New Delhi, India
B.Tech. – Information Technology